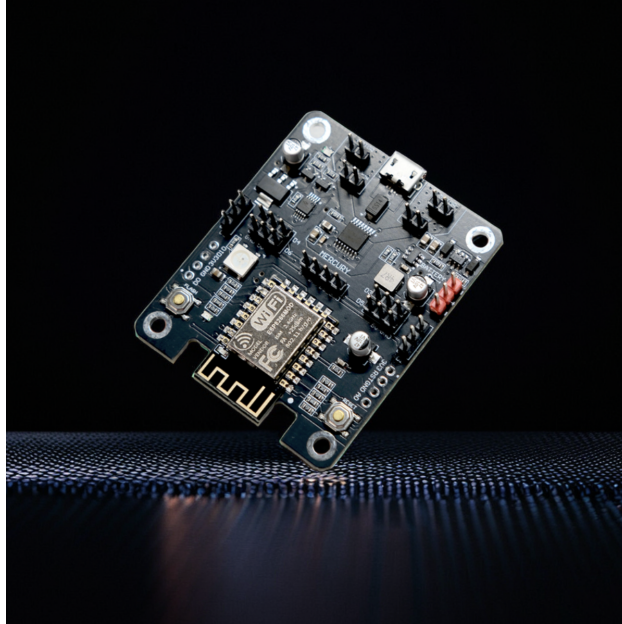
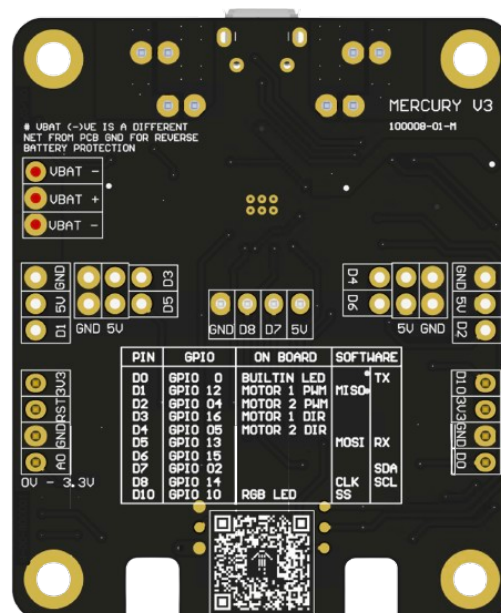
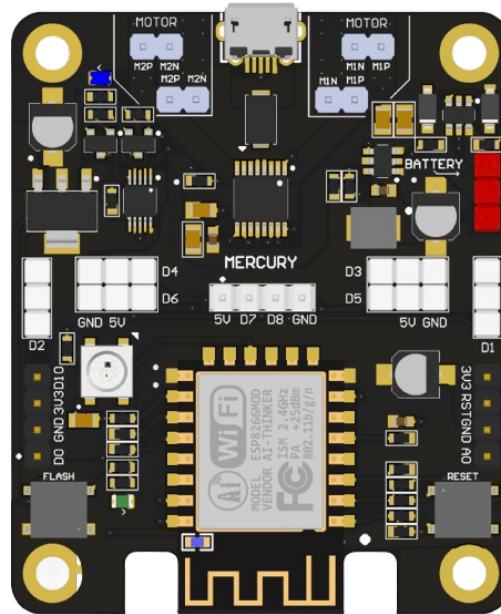


MERCURY V3 DEVELOPMENT BOARD DATASHEET

JULY 2025



The Mercury Board is a Make-in-India initiative by Ralio Technologies LLP, developed as an all-in-one microcontroller platform specifically tailored for robotics and STEM education. Based on ESP8266, Mercury integrates essential features such as an onboard motor driver, RGB LED, etc., and supports both Arduino and MicroPython programming. The latest version (v3) introduces major upgrades, including auto-flashing, on-board reverse battery protection, and improved power capacity. Designed for classroom learning and advanced prototyping, Mercury serves as the perfect brain for beginner to intermediate robotics and IoT projects.



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I. Key Features

- 3.3V operated
- External power I - 5V USB
- External power II - 12V (VBATT - Terminal)
- On-board reverse battery protection
- 3A - 12V to 5V Buck converter
- On-board WiFi (wireless protocol 802.11b/g/n)
- Supports software PWM
- Supports hardware/software I2C
- Supports software SPI
- Support software UART
- Analog Pin - 1
- Digital Pin - 10
- On-board WS2812 RGB LED - 1
- On-board motor driver - 2 channels (1A per channel)
- Support up to 6 servo motors (2.5A total maximum)
- Supported framework - Arduino, Micro-Python

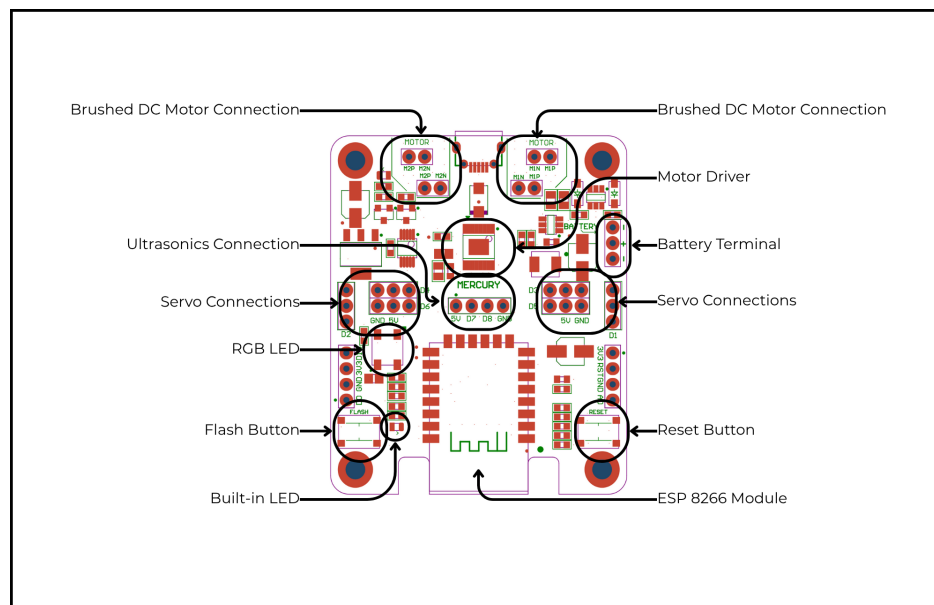


Fig. 1 Mercury v3 - Features

The Mercury Development Board is a versatile IoT and robotics platform designed around the ESP-12 module, powered by the ESP8266 Wi-Fi SoC from Espressif Systems. With built-in Wi-Fi connectivity and an onboard motor driver, Mercury is ideal for a wide range of applications, from driving four motors and ultrasonic sensors in a smart car to controlling up to six servo motors in a 6-DOF robotic arm.

Thanks to its integrated features, Mercury simplifies the development of IoT-based projects by offering seamless Wi-Fi support right out of the box. It also includes an onboard RGB LED and supports extension via WS2812 LED strips, making it easy to create vibrant visual feedback systems controlled from a single pin.

Whether you're building smart automation systems or hands-on robotics projects, Mercury offers a compact, plug-and-play solution tailored for both students and makers.

II. Voltage Rails

The Mercury Board provides multiple regulated voltage rails to simplify power distribution in projects:

- **VBAT (7V–12V):** The external power input rail, typically used to supply motors. It bypasses USB limitations and allows the Mercury Board to deliver higher current to connected actuators.
- **5V Rail:** Directly powered from a USB or external power source (via VBAT), a high-efficiency 3A buck converter steps down the VBAT voltage to provide a stable and reliable 5V output on the 5V rail. This rail supplies a stable 5V output suitable for sensors, modules, and logic-level devices.
- **3.3V Rail:** Internally regulated from 5V, this rail powers the ESP8266 and can also be used for other 3.3V components. It is recommended for low-power sensors and logic-level interfacing.

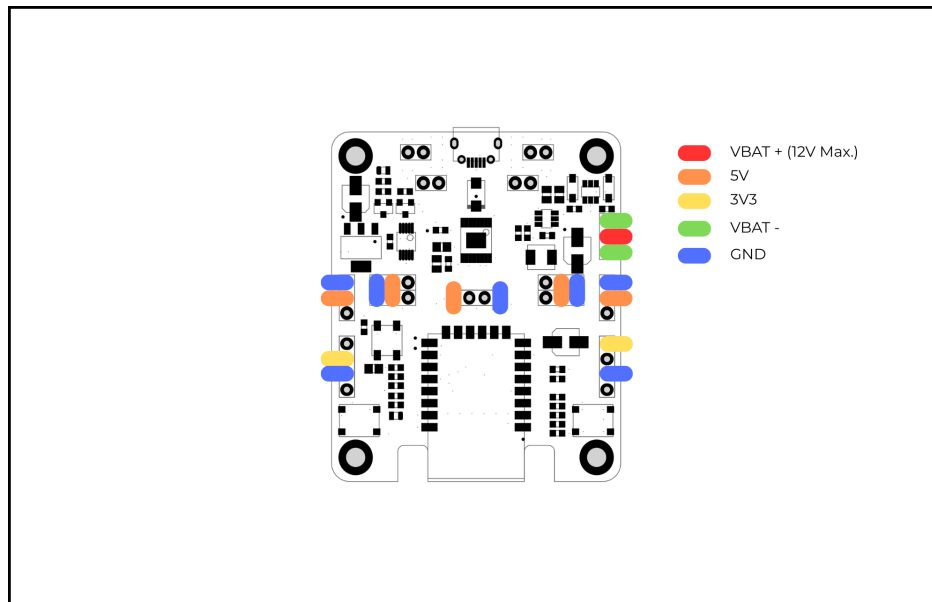


Fig. 2 Mercury v3 - Voltage Rails

All rails are clearly labeled and available through dedicated headers, enabling safe and flexible connectivity for classroom or prototyping environments.

III. Pin-out Description

Pin	GPIO	Hardware	Software	
A0	17	ADC	MISO (SPI)	TX (UART)
D0	0	BUILTIN LED		
D1	12	MOTOR 1 PWM		
D2	4	MOTOR 2 PWM		
D3	16	MOTOR 1 DIR		
D4	5	MOTOR 2 DIR	MOSI (SPI)	RX (UART)
D5	13			
D6	15			
D7	2		CLK (SPI)	SDA (I2C)
D8	14			SCL (I2C)
D10	10	RGB LED	SS (SPI)	

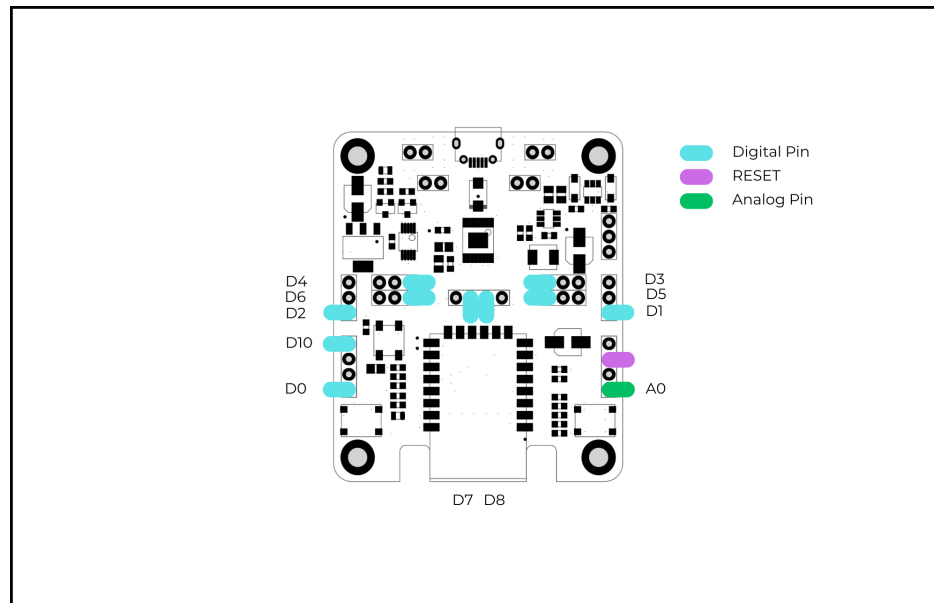


Fig. 3 Mercury v3 - Pinout

Pin	GPIO	Hardware Configuration
D0	0	PULL-UP
D6	15	PULL-DOWN
D7	2	PULL-UP
D8	14	PULL-UP

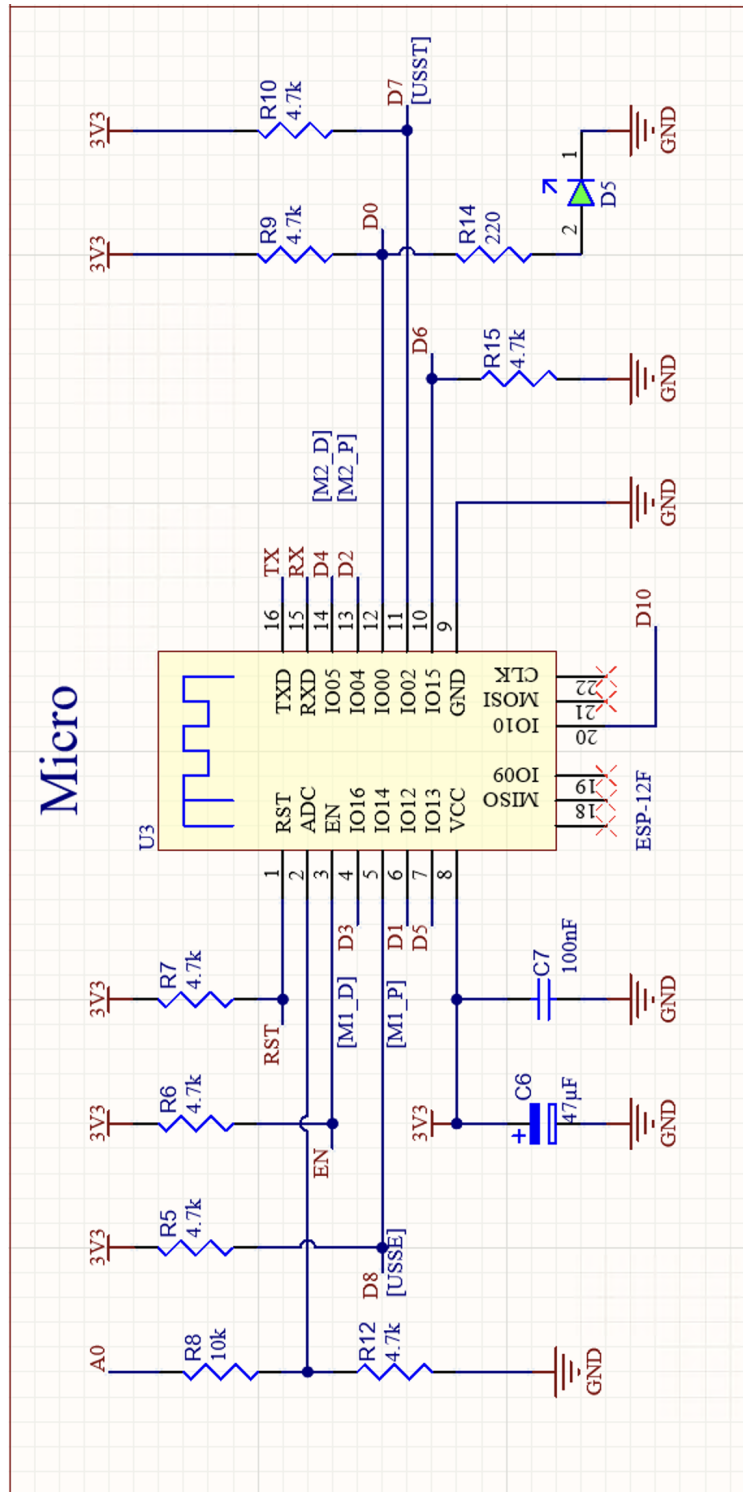


Fig. 4 Mercury v3 - Pinout

IV. Technical Specification

Board	
Name	Mercury
Version	3
SKU	100008-01-M
Micro-controller	
Name	ESP8266EX
Module	ESP12-F
Peripheral Interface	
Digital I/O	10
ADC I/O	1 (Input: 0V to 3.3V)
PWM	Yes (Software)
Connectivity and Communication	
UART	Yes (Software)
SPI	Yes (Software)
I2C	Yes (Hardware/Software)
USB	Micro-USB
WiFi	2.4GHz
Electrical	
GPIO Voltage	3.3V (Nominal)
GPIO Voltage Absolute Max.	3.6V
Source Current per GPIO	10mA (Nominal)
Source Current per GPIO Absolute Max.	15mA
Sink Current per GPIO	15mA (Nominal)
Sink Current per GPIO Absolute Max.	20mA
External Battery Voltage	7V to 12V (Nominal)
External Battery Voltage Absolute Max.	15V
External Battery Current Absolute Max.	3A
Additional Resources: ESP8266EX Datasheet	
Thermals	
Operational Temperature range	-20C to 85C

V. Mechanical

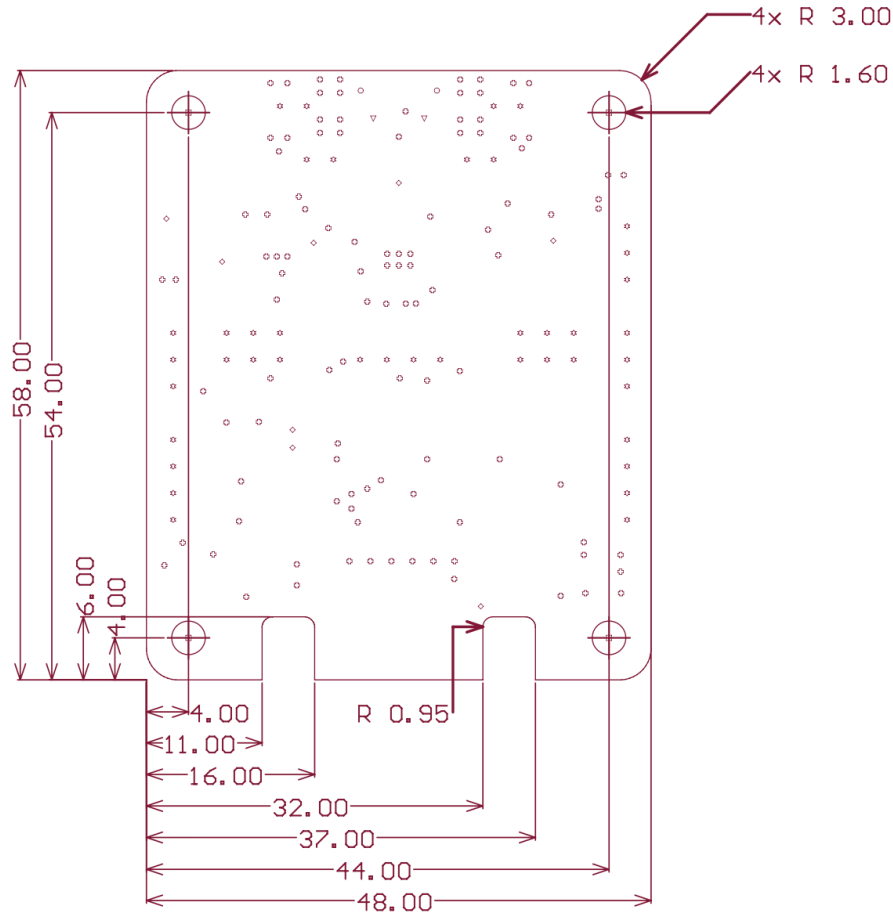


Fig. 5 Mercury v3 - Dimension [Units: mm]

VI. Arduino® Support

Mercury Board is supported on Arduino® platform. Detailed instructions can be found on this Github page: <https://github.com/raliotech/products/blob/master/boards/mercury/README.md>

Board Package JSON File:

https://raw.githubusercontent.com/raliotech/products/master/core/package_ralio_index.json

Additional ESP8266 Firmware Support:

<https://github.com/esp8266/Arduino>

— END —